

# Electrical Topology Diagram (Implementation v6)

As-of date: 2026-07-19

Purpose: provide a complete, install-level electrical topology for the current build scope, including all major electrical components, fuse IDs, fuse housings, planned wire gauges, and estimated one-way run lengths for procurement planning.

Related docs:

- Canonical electrical/system baseline: docs/core/SYSTEMS.md
- Detailed fuse matrix: docs/implementation/ELECTRICAL\_fuse\_schedule.md
- Battery and trunk recalculation record: docs/studies/ELECTRICAL\_battery\_fuse\_wire\_recalc\_2026-02-18.md
- Decisions and unresolved items: docs/core/TRACKING.md
- Procurement source of truth: bom/bom\_estimated\_items.csv

## Sweep Outcomes Included In This Revision

- Keeps alternator charging architecture on the dedicated 48V secondary alternator path ( Mechman + WS500 + APM-48 baseline); obsolete pre-Mechman charger paths are removed from primary topology.
- Keeps Lynx Slot 3 branch to alternator input with F-04 150A ( 58V/80V MEGA).
- Removes obsolete pre-Mechman engine-bay fuse/conductor placeholders from active architecture.
- Clarifies the confirmed PH-VAN harness as one 15A fused combined power/positive-sense red lead ( F-12/F-13-PHVAN ) and locks the separate Wakespeed 500A/50mV shunt into alternator B+ , with one bank-voltage-rated 5A fuse in each purple/grey sense lead immediately at the shunt.
- Adds Ford Upfitter #3 -> F-15 -> WS500 brown ignition manual alternator-control path.
- Adds the detailed Mechman/WS500/APM-48 install guide as the shop reference for staged installation, first-run checks, and load-dump/shutdown handling.
- Defines APM-48 as a parallel surge clamp at the alternator rather than a series charge-current device.
- Added explicit fuse-holder/housing definitions for every fuse family ( Class T , Lynx MEGA , inline MIDI/ANL/AMI , PV gPV , and AT0/ATC ).
- Added conductor schedule across 48V , 12V , PV, and AC segments with explicit assumptions.
- Updated 12V topology to a shared 12V junction fed by an Orion-Tr Smart 48/12-30 charger and a 12V 100Ah buffer battery branch, with F-11 source fuse plus SW-12V-BATT manual isolation.
- Added a full-circuit estimated run-length validation pass ( C-01 through C-40 ) and purchase-ready wire rollup totals.

## Current Commissioning Snapshot ( 2026-08-02 )

- 48V bus live-tested at 55.5V throughout the system, including at the MultiPlus.
- MultiPlus-II inverter mode tested with inverter light on, slight normal hum, and no reported errors.
- SmartShunt and Orion-Tr Smart connected in VictronConnect.
- Cerbo GX access point/remote-console workflow active; Cerbo is powered from a small inline fused 48V feed and connected to MultiPlus via VE.Bus RJ45.
- Short AC-in shore-charge test passed at household-source current limits: about 1294W shore input and about 54.3V x 21.6A battery charging in bulk.
- MultiPlus LiFePO4 charge profile is programmed/owner-verified by supervised first-battery behavior. Owner reports the Orion/ 12V buffer path now charges and operates correctly. Hold open: permanent-path shore dead checks/first-use acceptance, full-bank charge/rest/match/parallel closeout, AC-out branch/GFCI, alternator commissioning, Cerbo hard-mounting, and final strain-relief/abrasion-control.

## Current Physical Installation Snapshot ( 2026-08-03 )

- The electrical module is hard-mounted through the finished Lonseal/plywood floor into registered truck-bed hardpoints and tied into the hard-mounted Bench/Galley extrusion structure. Owner reports the integrated assembly is extremely stiff; remaining mobile restraint is battery/cooler capture plus terminal/cable protection.
- The driver-rear shore inlet and exterior-to-module cable route are installed. One accessible, enclosed, conductor/gauge-rated three-wire L/N/PE splice into the AC-input side remains before dead checks and controlled energization.
- All three batteries' positive and negative 2/0 AWG branch cables are cut, lugged, heat-shrunk, and landed at the battery-side busbars. Battery 1 completed the corrected isolated charge cycle, Battery 2 resumed and reached absorption on 2026-08-03 , and Battery 3 remains pending; keep all three isolated until their rested voltages are matched within 0.1V before paralleling.
- The 12V buffer-battery/Orion branch is owner-reported operational. Keep its negative direct to the fuse-panel main negative stud, not through SW-12V-BATT ; the positive path remains F-11 -> SW-12V-BATT -> panel main + .

- Owner report 2026-08-03 : the ICECO appliance lead is landed on 12V-02 , protected by a 15A blade fuse, and switched. Its appliance pigtail is 16 AWG ; exposed positive switch terminals still require individual insulation and temporary hard mounting before normal powered work. Verify connector polarity and voltage drop before appliance acceptance.
- Owner report 2026-08-05 : the pump, ICECO/fridge feed, and KUS sender circuit are wired; three separate 12 AWG / 15A USB-PD branches are routed ( 1x Desk and 2x Galley); and the Desk and Galley GFCIs are wired on separate breakers. Commissioning remains open for exact fuse/slot labels, polarity/load tests, KUS configuration/sanity check, AC LINE/LOAD/PE proof, and GFCI trip/reset. The third PD branch's actual source slot must be audited and labeled rather than inferred from this document.

## Orion fuse discriminator — one input fuse only

- Lynx Slot 4 / F-05 : **Orion 48V input positive**, one verified 40A MEGA ( 58VDC minimum under the locked 56.8V charge ceiling; Victron CIP138040020 40A/80V is the replacement fallback) feeding the existing 6 AWG directly. This is the final single input fuse.
- F-06 : **retired standalone Orion input-fuse position**. Do not install MIDI, FKS/ATO, DIN, or a second fuse after Slot 4.
- Lynx Slot 2 / F-03 : **MPPT branch**, 60A/80V MEGA. If the physically X-marked/misrated Lynx fuse is in Slot 2, replace it with the purchased Victron 60A/80V MEGA; it is not the Orion input fuse.
- F-07 : **Orion 12V output positive**, 60A/80V MEGA in the separate Victron holder near the Orion.
- F-11 : **12V buffer-battery positive**, 100A ANL near the battery, upstream of SW-12V-BATT .
- No 32V -rated fuse is acceptable on any energized 48V branch. Identify the physical X-marked fuse by Lynx slot before replacing or discarding it.

## Length Estimation Defaults Used In This Pass

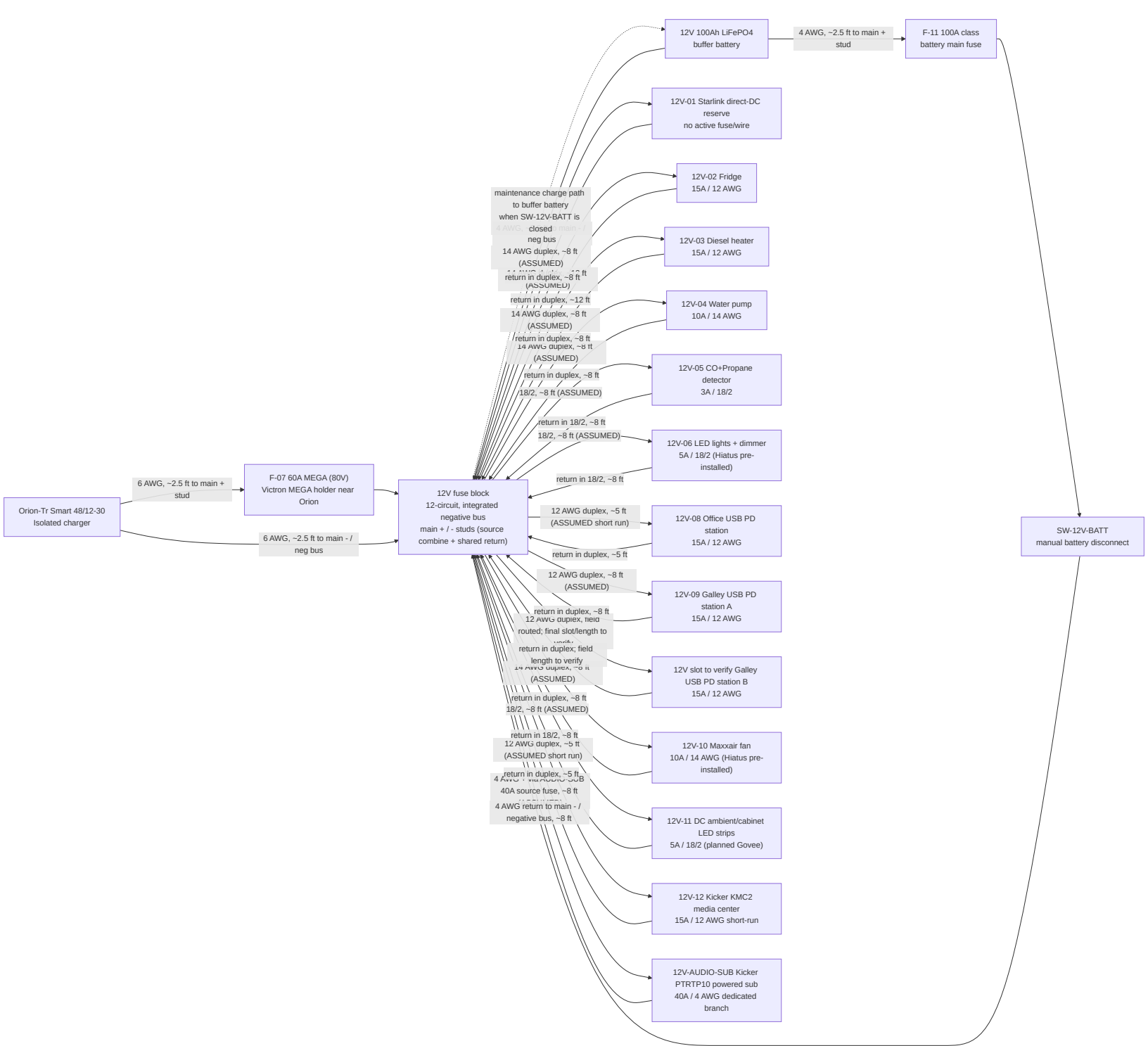
1. Cabinet internal interconnect default: 2.5 ft one-way ( ASSUMED ).
2. Cabinet-to-near load branch default: 8 ft one-way ( ASSUMED ).
3. Cabinet-to-far load branch default: 12 ft one-way ( ASSUMED ).
4. AC branch to receptacle chain default: 15 ft one-way per branch leg ( ASSUMED ).
5. Policy lock: use the smallest gauge that meets current and voltage-drop targets; do not auto-upsize, but flag warnings when margin is tight.
6. Parallel battery bank lock: keep each battery's **total positive + negative path resistance** similar. Equal positive-only length is not required if the positive-length differences are offset by negative-length differences; do not add unnecessary 2/0 cable coils solely for cosmetic equality.

## Battery Fuse/Wire Recalculation Basis (2026-02-18 + 2026-03-19 sync)

- Scope in this pass is limited to battery-side and major 48V trunk paths ( C-01 through C-15 ).
- Provisional battery listing inputs used: 51.2V 100Ah ,  $\leq 200A$  current limit per battery.
- Conservative sizing factors used in this pass:
  1. Parallel-sharing factor  $K_{share} = 1.5$
  2. Continuous margin factor  $K_{cont} = 1.25$
- Current envelope used for battery-discharge branch sizing:  $I_{total} = F-02 + F-05 = 125A + 40A = 165A$  .
- Per-battery design current:  $I_{batt\_design} = (165A / 3) * 1.5 = 82.5A$  .
- Continuous-adjusted minimum battery branch fuse threshold:  $82.5A * 1.25 = 103.1A$  .
- Provisional battery branch fuse selection: F-01A/B/C = 200A Class T , constrained by the provisional battery  $\leq 200A$  current-limit listing.
- Final lock gate: validate true 51.2V battery datasheet/manual current and terminal limits before permanent fuse lock; if lower limits are confirmed, move to 175A .
- Cable procurement remains estimate-based until CAD/field run lengths are frozen. This pass sets a no-padding 2/0 estimate baseline of 77.5 ft total ( 42.5 ft red, 35.0 ft black), including the dedicated alternator migration path.

## Complete Power Topology (48V Core + Charge Sources)



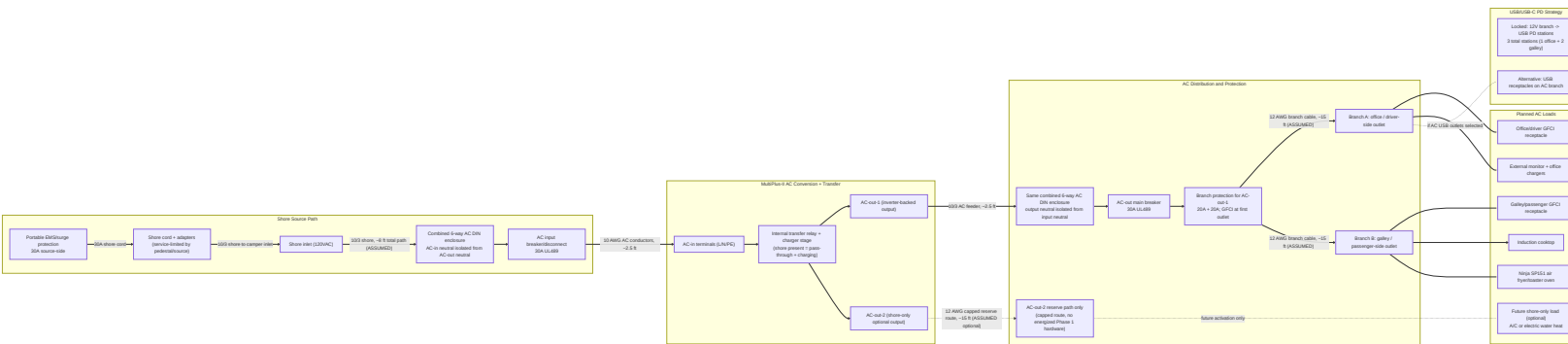


## 12V Operating Intent (Locked)

- Orion-Tr Smart 48/12-30 is the primary 12V charger/feed source.
- SW-12V-BATT is **normally closed** in operation; open is service/isolation mode only.
- With SW-12V-BATT closed, Orion output at the fuse-block main + stud maintains/charges the 12V buffer battery through the shared-junction path.
- With SW-12V-BATT open, the buffer battery is isolated from the main + stud, but the fuse-panel bus can remain powered from the Orion while the Orion is enabled. For service isolation, turn active loads off, disable the Orion and confirm off/not charging, then open SW-12V-BATT and meter the bus before work.
- Replace the Orion's always-on L-H jumper with a maintained SPST dry-contact remote switch. Startup closes SW-12V-BATT before enabling the Orion; normal shutdown disables the Orion before opening the battery switch.
- The buffer battery remains in the active operating path during normal use and is intended to absorb transients/peaks on the 12V rail.
- The fuse block is the 12V junction device in this baseline: main + stud is the source-combine point, and the integrated negative bus/main - is the shared return point.
- Do not solder-splice high-current source conductors; terminate with crimped lugs on rated studs/junction hardware.

- Hiatus pre-installed 12V branches are tracked here as 12V-06 (factory LED lights with dimmer) and 12V-10 (Maxxair fan); verify final installed branch labeling during electrical audit.
- Planned Govee/ambient strip lighting is a separate branch ( 12V-11 ) and not part of the Hiatus factory 12V-06 lighting circuit.
- DC-first lighting intent: keep ambient/cabinet strip lighting on 12V-11 so nighttime lighting does not require inverter operation.
- Camper audio intent: Kicker 46KMC2 source unit uses a 15A 12V-12 branch from the fuse block, while Kicker 49PTRTP10 powered sub uses a separate 40A fused 4 AWG branch from the 12V source/junction rather than a normal blade-fuse panel branch. Bass peaks may exceed the Orion 30A continuous feed and are supported by the 12V buffer battery for short durations.

## AC Path Topology (Shore + Inverter Output, Full Hierarchy)



## AC Operating Behavior (Expected)

- Shore present: MultiPlus transfer relay closes, AC-in is passed to AC-out paths, and charger stage charges the 48V bank.
- Shore absent: MultiPlus transfers to inverter mode and powers AC-out-1 from battery; AC-out-2 drops by design.
- AC-in hardware is 30A ( source/adapters -> portable EMS -> shore cord -> L5-30 inlet -> 30A breaker -> 10 AWG AC-in conductors ); set MultiPlus input current limit to actual source ( 15A , 20A , or 30A ) to avoid pedestal/source breaker trips.
- Initial battery charging may use AC-in-only mode with AC-out branch loads disconnected.

## AC Safety/Protection Chain (What Must Exist)

- Upstream shore protection chain before MultiPlus AC-in: shore source/adapters -> portable EMS -> shore cord -> shore inlet -> 30A AC input breaker/disconnect.
- AC-out protection chain: MultiPlus AC-out-1 -> 10/3 feeder -> 30A AC-out main breaker -> 20A branch breakers -> GFCI receptacles.
- Single-enclosure DIN architecture with isolated AC-in and AC-out neutral paths and no neutral mixing.
- Continuous equipment grounding path from shore inlet through MultiPlus and branch circuits. The MultiPlus external M6 PE lug gets at least 4 mm<sup>2</sup> , selected as 10 AWG green stranded copper, to a verified truck-chassis bond point; the aluminum shell gets a separate corrosion-compatible bond into the same equipment-ground network and is not the only PE path.
- Neutral/ground handling follows the MultiPlus internal relay: AC-out neutral bonds to chassis in inverter mode and opens that bond when shore is accepted. Do not add an always-bonded downstream neutral-ground bond in branch receptacle wiring.
- Keep PE and DC return roles separate: no MultiPlus case/PE jumper to Lynx or 12V negative. Confirm whether the Mechman is isolated- or case-grounded and prove no chassis path bypasses the SmartShunt before alternator commissioning.

## AC Reference Basis (Manufacturer Guidance)

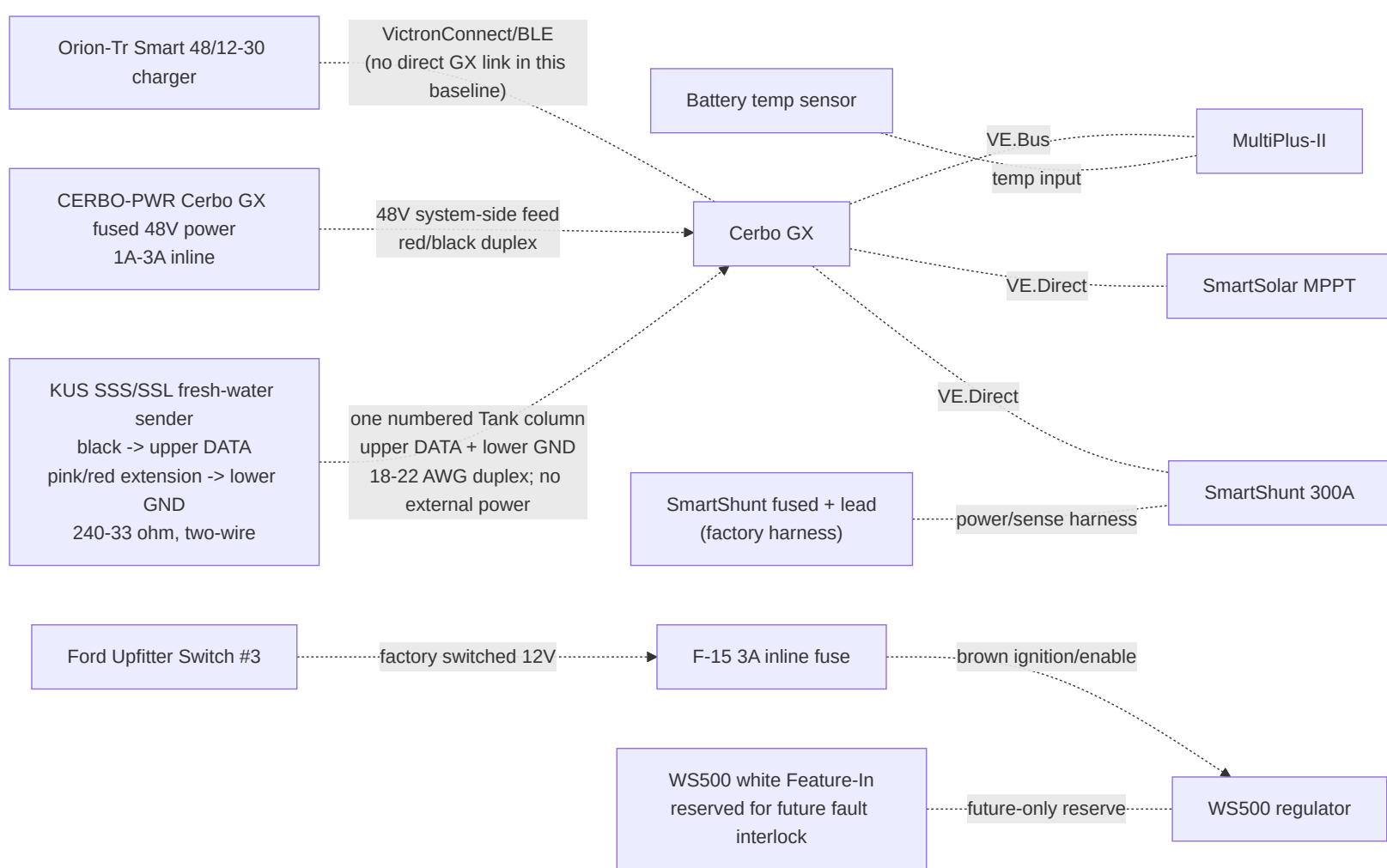
- Victron MultiPlus-II 120V installation guidance ( AC-in breaker sizing, UL943-class residual-current protection on outputs, and AC-out-2 shore-only behavior): [https://www.victronenergy.com/media/pg/MultiPlus-II\\_120V/en/installation.html](https://www.victronenergy.com/media/pg/MultiPlus-II_120V/en/installation.html)
- Victron MultiPlus-II datasheet ( 48/3000/35-50 baseline model reference): <https://www.victronenergy.com/upload/documents/Datasheet-MultiPlus-II-inverter-charger-120V-EN.pdf>

## AC/USB Baseline Locked For BOM

- Shore interface: 30A RV-style inlet baseline with adapter kit for 15A / 20A hookups.
- AC input protection: portable EMS + combined DIN enclosure + 30A AC input breaker/disconnect upstream of MultiPlus AC-in.
- AC-out-1 distribution: 30A AC-out main plus two protected 20A branches with GFCI-at-first-outlet strategy.
- Receptacle plan: 2 first-in-chain 120V GFCI receptacles remain required, one per active branch: Branch A = office/driver side, Branch B = galley/passenger side. Owner reopened a small number of downstream receptacles on 2026-08-05 ; final count/locations remain field-gated and must continue from each first GFCI's LOAD terminals with 12 AWG conductors, accessible listed boxes, and full-chain trip testing.
- USB charging plan: 3 DC-fed USB PD station assemblies on separate 12V branches ( 1 office, 2 Galley), each standardized to 15A / 12 AWG . The third branch's actual fuse-panel slot/label is a field-audit item.

- AC-out-2 remains reserve-only in Phase 1 (labeled capped route; no energized branch hardware procured).

## Monitoring and Control Topology



## Fuse and Switch Housing Map (Where Each Item Is Physically Housed)

| Item ID | Item value/type                                                                  | Housing method                                 | Location                                                  |
|---------|----------------------------------------------------------------------------------|------------------------------------------------|-----------------------------------------------------------|
| F-01A   | 200A Class T (provisional)                                                       | Blue Sea Class T fuse block (110A-200A family) | Battery compartment near Battery A +                      |
| F-01B   | 200A Class T (provisional)                                                       | Blue Sea Class T fuse block (110A-200A family) | Battery compartment near Battery B +                      |
| F-01C   | 200A Class T (provisional)                                                       | Blue Sea Class T fuse block (110A-200A family) | Battery compartment near Battery C +                      |
| F-02    | 125A MEGA                                                                        | Lynx integrated slot holder                    | Lynx Slot 1                                               |
| F-03    | 60A MEGA (80V Victron replacement stock)                                         | Lynx integrated slot holder                    | Lynx Slot 2                                               |
| F-04    | 150A MEGA                                                                        | Lynx integrated slot holder                    | Lynx Slot 3 (dedicated alternator branch)                 |
| F-05    | 40A MEGA, body-marked >=58VDC; Victron CIP138040020 40A/80V replacement fallback | Lynx integrated slot holder                    | Lynx Slot 4; Orion 48v input feeder                       |
| F-06    | Not installed / retired standalone input-fuse position                           | No holder                                      | MIDI/FKS/DIN concepts superseded; do not stack after F-05 |

|                          |                                                                        |                                                                                                      |                                                                                                                                                           |
|--------------------------|------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|
| F-07                     | 60A MEGA (80V Victron replacement stock)                               | Victron MEGA fuse holder                                                                             | Electrical cabinet at Orion 12V + source end                                                                                                              |
| F-09A/B/C                | 15A gPV each                                                           | 10x38 touch-safe fuse holders in PV combiner                                                         | Roof-entry combiner enclosure                                                                                                                             |
| F-10                     | Per branch (ATO/ATC)                                                   | Integrated blade sockets in generic 12V fuse block                                                   | Electrical cabinet                                                                                                                                        |
| AUDIO-HU / 12V-12        | 15A source/head-unit branch; KMC2 harness also contains a 15A ATM fuse | 12V fuse block branch plus KMC2 harness fuse                                                         | Electrical cabinet to driver-side DC shelf/source face                                                                                                    |
| AUDIO-SUB                | 40A external fuse for Kicker PTRTP10 powered sub branch                | Inline/MRBF/AFS/ANL-class holder matched to selected 4 AWG kit; use 40A, not a generic 100A kit fuse | Within about 18 in of the 12V source takeoff feeding the powered sub                                                                                      |
| F-11                     | 100A class (12V buffer battery main)                                   | Sealed inline MIDI/AMI/ANL holder                                                                    | Within ~`7"`` of 12V buffer battery positive post                                                                                                         |
| F-12/F-13-PHVAN          | 15A WS500 combined regulator-power / positive-sense fuse               | Sealed inline holder rated above actual 48V bank maximum                                             | At the house/main positive bus feeding the short PH-VAN red lead; do not extend                                                                           |
| CERB0-PWR                | 1A-3A Cerbo GX power fuse                                              | Small inline holder rated for the 48V bank maximum                                                   | Electrical cabinet near 48V system-positive takeoff; system side of disconnect preferred for bench shutdown                                               |
| WS500 current-sense pair | No fuse; purple/grey high/low sense pair to shunt/current-sense point  | Twist pair if extended; route away from noise                                                        | Shunt/current-sense source point                                                                                                                          |
| F-15                     | 3A WS500 ignition/enable control fuse                                  | Sealed inline ATC/ATO holder; 12V control circuit                                                    | Near Ford upfitter blunt-cut wire / WS500 control-wire handoff                                                                                            |
| SW-12V-BATT              | Manual battery disconnect switch                                       | Sealed rotary DC switch body                                                                         | Electrical cabinet near 12V fuse-block main + stud for service access                                                                                     |
| OEM-SHUNT                | External Victron-supplied inline fuse in red SmartShunt Vbatt+ cable   | Inline holder in supplied red cable                                                                  | Prefer battery-side positive if SOC continuity is desired with the main disconnect open; system side is acceptable for zero disconnect-off parasitic draw |

Retired from active architecture:

- Obsolete pre-Mechman charger/fuse paths are removed from active board layout and labels.

## Conductor Schedule (Start-to-Finish)

| Segment ID | Circuit segment      | Nominal voltage | Current basis                | Overcurrent protection | Planned wire gauge | Estimated one-way length (this pass)                 |
|------------|----------------------|-----------------|------------------------------|------------------------|--------------------|------------------------------------------------------|
| C-01       | Battery A + -> F-01A | 48V             | Battery branch, fuse-limited | F-01A 200A provisional | 2/0 AWG            | 2.5 ft planning placeholder; balance total loop path |
| C-02       | Battery B + -> F-01B | 48V             | Battery branch, fuse-limited | F-01B 200A provisional | 2/0 AWG            | 2.5 ft planning placeholder; balance total loop path |

|           |                                                                           |     |                                                                                                                |                                        |                                                         |                                                      |
|-----------|---------------------------------------------------------------------------|-----|----------------------------------------------------------------------------------------------------------------|----------------------------------------|---------------------------------------------------------|------------------------------------------------------|
| C-02C     | Battery C + -> F-01C                                                      | 48V | Battery branch, fuse-limited                                                                                   | F-01C 200A provisional                 | 2/0 AWG                                                 | 2.5 ft planning placeholder; balance total loop path |
| C-03      | Class T outputs -> battery-side 48V + busbar -> disconnect input          | 48V | Combined trunk current                                                                                         | F-01A/B/C                              | 2/0 AWG each branch                                     | 2.5 ft each branch (ASSUMED, 4 conductors in rollup) |
| C-04      | Disconnect output -> Lynx + bus                                           | 48V | Aggregate discharge design current (165A = F-02 125A + Orion F-05 40A)                                         | Upstream Class T fuses                 | 2/0 AWG                                                 | 2.5 ft (ASSUMED)                                     |
| C-05      | Battery negatives -> battery-side 48V - busbar -> SmartShunt battery side | 48V | Mixed-path rollup: 3x battery-negative branches at 82.5A design each + NEGBUS_TO_SHUNT trunk at 165A aggregate | N/A (main negative path)               | 2/0 AWG each branch                                     | 2.5 ft each branch (ASSUMED, 4 conductors in rollup) |
| C-06      | SmartShunt load side -> Lynx - bus                                        | 48V | Aggregate return current                                                                                       | N/A                                    | 2/0 AWG                                                 | 2.5 ft (ASSUMED)                                     |
| C-06A     | Lynx positive tap -> SmartShunt positive sense/power lead                 | 48V | Shunt electronics supply (very low current)                                                                    | Factory inline fuse in OEM harness     | OEM harness lead                                        | 2.5 ft (ASSUMED)                                     |
| C-07      | Lynx Slot 1 (F-02) -> MultiPlus DC+                                       | 48V | Inverter branch, fuse-limited                                                                                  | F-02 125A                              | 2/0 AWG (manual minimum AWG 1 on short runs)            | 2.5 ft (ASSUMED)                                     |
| C-08      | MultiPlus DC- -> Lynx - bus                                               | 48V | Inverter return current                                                                                        | F-02 protects paired positive          | 2/0 AWG                                                 | 2.5 ft (ASSUMED)                                     |
| C-09      | MPPT BAT+ -> Lynx Slot 2 (F-03)                                           | 48V | Controller output (45A max)                                                                                    | F-03 60A/80V Victron row 188           | 6 AWG                                                   | 2.5 ft (ASSUMED)                                     |
| C-10      | MPPT BAT- -> Lynx - bus                                                   | 48V | Controller return current                                                                                      | F-03 protects paired positive          | 6 AWG                                                   | 2.5 ft (ASSUMED)                                     |
| C-11      | Secondary alternator B+ -> APM-48 -> Lynx Slot 3 (F-04)                   | 48V | Alternator branch design current                                                                               | F-04 150A                              | 2/0 AWG                                                 | 20 ft (ASSUMED, one-way)                             |
| C-12      | Secondary alternator B- -> Lynx - bus (dedicated return)                  | 48V | Alternator branch return current                                                                               | F-04 paired                            | 2/0 AWG                                                 | 20 ft (ASSUMED, one-way)                             |
| C-13/C-14 | Lynx Slot 4 (F-05) -> Orion 48V + input                                   | 48V | Orion feeder, fuse-limited                                                                                     | F-05 40A MEGA, >=58VDC                 | Existing 6 AWG direct; 10 AWG adequate if ever replaced | 2.5 ft (ASSUMED)                                     |
| C-15      | Orion 48V - input -> Lynx - bus                                           | 48V | Orion input return current                                                                                     | Orion input positive protection paired | 6 AWG                                                   | 2.5 ft (ASSUMED)                                     |



|       |                                                                           |                        |                                                                               |                                                                  |                                               |                                                                                |
|-------|---------------------------------------------------------------------------|------------------------|-------------------------------------------------------------------------------|------------------------------------------------------------------|-----------------------------------------------|--------------------------------------------------------------------------------|
| C-18  | Orion 12V + -> F-07 -> 12V fuse block main + stud                         | 12V                    | Charger output path (30A continuous, 60A fuse)                                | F-07 60A                                                         | 6 AWG planned (8 AWG minimum per Orion table) | 2.5 ft (ASSUMED)                                                               |
| C-19  | Orion 12V - -> 12V fuse block integrated - bus / main - stud              | 12V                    | Charger output return                                                         | F-07 protects paired positive                                    | 6 AWG                                         | 2.5 ft (ASSUMED)                                                               |
| C-19A | 12V buffer battery + -> F-11 -> SW-12V-BATT -> 12V fuse block main + stud | 12V                    | Buffer source path and service isolation path                                 | F-11 100A class                                                  | 4 AWG planned                                 | 2.5 ft (ASSUMED)                                                               |
| C-19B | 12V buffer battery - -> 12V fuse block integrated - bus / main - stud     | 12V                    | Buffer battery return path                                                    | N/A (paired with C-19A)                                          | 4 AWG planned                                 | 2.5 ft (ASSUMED)                                                               |
| C-20  | Reserved 12V panel -> future Starlink direct-DC conversion                | 12V                    | Not active; Standard 4 X remains on its supplied AC power path                | No fuse installed until exact converter/input is selected        | No conductor committed                        | Pull string/conduit only if useful while access is open                        |
| C-21  | 12V panel -> Fridge                                                       | 12V                    | Branch load                                                                   | F-10 15A                                                         | 12 AWG duplex                                 | 12 ft (ASSUMED, far-load branch; upsize avoids the prior voltage-drop warning) |
| C-22  | 12V panel -> LF Bros diesel-heater harness                                | 12V                    | Startup/cooldown branch; keep energized through controller-commanded shutdown | F-10 15A (retain any supplied harness fuse; verify coordination) | 12 AWG duplex                                 | 8 ft (ASSUMED; measure before final cable landing)                             |
| C-23  | 12V panel -> Water pump                                                   | 12V                    | Branch load                                                                   | F-10 10A                                                         | 14 AWG duplex                                 | 8 ft (ASSUMED, near-load branch)                                               |
| C-24  | 12V panel -> CO + propane detector                                        | 12V                    | Branch load                                                                   | F-10 3A                                                          | 18/2                                          | 8 ft (ASSUMED, near-load branch)                                               |
| C-25  | 12V panel -> LED lights + dimmer (Hiatus pre-installed)                   | 12V                    | Branch load                                                                   | F-10 5A                                                          | 18/2                                          | 8 ft (ASSUMED, near-load branch)                                               |
| C-26  | 48V system positive/negative -> CERBO-PWR -> Cerbo GX power input         | 48V                    | Cerbo electronics feed (~3W)                                                  | CERBO-PWR 1A-3A inline fuse                                      | 18 AWG red/black duplex acceptable            | 2.5 ft (ASSUMED, cabinet internal)                                             |
| C-27  | PV strings -> F-09 combiner -> MPPT PV input                              | PV string voltage (3S) | String current + combiner output current                                      | F-09A/B/C 15A each                                               | 10 AWG PV wire                                | 12 ft trunk + 3x8 ft string legs (ASSUMED)                                     |
| C-28  | Shore source/adapters ->                                                  | 120VAC                 | Source-limited shore current (adapter-                                        | 30A AC input breaker/disconnect                                  | 10/3 shore feed to                            | 8 ft (ASSUMED)                                                                 |

|      |                                                                                           |                   |                                                                                |                                                    |                                                   |                                                                                         |
|------|-------------------------------------------------------------------------------------------|-------------------|--------------------------------------------------------------------------------|----------------------------------------------------|---------------------------------------------------|-----------------------------------------------------------------------------------------|
|      | portable EMS -> shore cord -> shore inlet -> combined AC DIN enclosure / AC input breaker |                   | constrained at source)                                                         | baseline with source-current-limit settings policy | inlet/AC-in area                                  |                                                                                         |
| C-29 | AC input breaker/disconnect -> MultiPlus AC-in                                            | 120VAC            | MultiPlus AC input current (30A hardware basis)                                | Upstream 30A AC breaker/disconnect (C-28)          | 10 AWG stranded AC conductors                     | 2.5 ft (ASSUMED, cabinet internal)                                                      |
| C-30 | MultiPlus AC-out-1 -> combined AC DIN enclosure / AC-out main breaker                     | 120VAC            | Inverter-backed AC-out feeder current (30A system cap)                         | 30A AC-out main breaker                            | 10/3 stranded AC feeder                           | 2.5 ft (ASSUMED, cabinet internal)                                                      |
| C-31 | Branch A -> office/driver GFCI receptacle                                                 | 120VAC            | Branch load (office monitor/chargers/general outlet use)                       | 20A branch breaker + GFCI receptacle               | 12/3 stranded AC branch cable                     | 15 ft (ASSUMED, branch leg default)                                                     |
| C-32 | Branch B -> galley/passenger GFCI receptacle                                              | 120VAC            | Branch load (galley/general high-draw outlet; induction/Ninja SP151 sequenced) | 20A branch breaker + GFCI receptacle               | 12/3 stranded AC branch cable                     | 15 ft (ASSUMED, branch leg default)                                                     |
| C-33 | MultiPlus AC-out-2 (reserve-only) -> capped route for future shore-only branch            | 120VAC            | N/A in Phase 1 (route reserved only)                                           | N/A in Phase 1 (no energized branch hardware)      | 12 AWG stranded AC conductors (reserve path only) | 15 ft (ASSUMED, reserve route default)                                                  |
| C-34 | 12V panel -> USB PD station branch (office zone)                                          | 12V               | One Acegoo 118W station                                                        | F-10 branch fuse (15A)                             | 12 AWG duplex baseline                            | 5 ft (ASSUMED, short-run requirement)                                                   |
| C-35 | 12V panel -> USB PD station branch (galley zone)                                          | 12V               | Galley charging branch (65W class USB-C plus USB-A/C loads)                    | F-10 branch fuse (15A)                             | 12 AWG duplex baseline                            | 8 ft (ASSUMED; standardized with the office PD/fridge rough-in for voltage-drop margin) |
| C-36 | 12V panel -> Maxxair fan (Hiatus pre-installed)                                           | 12V               | Roof ventilation branch                                                        | F-10 branch fuse (10A)                             | 14 AWG duplex baseline                            | 8 ft (ASSUMED, near-load branch)                                                        |
| C-37 | 12V panel -> DC ambient/cabinet LED strips (planned Govee)                                | 12V               | Branch load                                                                    | F-10 branch fuse (5A)                              | 18/2 baseline                                     | 8 ft (ASSUMED, near-load branch)                                                        |
| C-38 | House/main positive bus -> WS500 PH-VAN short red lead                                    | 48V bank          | Combined regulator-power / positive-voltage-sense feed                         | F-12/F-13-PHVAN (15A)                              | Short harness lead; do not extend                 | Local at house bus                                                                      |
| C-39 | Retired separate WS500 positive-sense conductor                                           | N/A               | Not installed with confirmed PH-VAN harness                                    | None                                               | N/A                                               | N/A                                                                                     |
| C-40 | WS500 current-sense high/low pair to selected                                             | low-current sense | Regulator current feedback                                                     | No fuse per current Wakespeed manual;              | Harness lead                                      | 8 ft (ASSUMED)                                                                          |

|         |                                                                                                              |                                               |                                                              |                                                                   |                                                                              |                                                                   |
|---------|--------------------------------------------------------------------------------------------------------------|-----------------------------------------------|--------------------------------------------------------------|-------------------------------------------------------------------|------------------------------------------------------------------------------|-------------------------------------------------------------------|
|         | shunt/current-sense point                                                                                    |                                               |                                                              | twist pair if extended                                            |                                                                              |                                                                   |
| C-41    | Ford Upfitter #3 output -> F-15 -> WS500 brown ignition/enable wire                                          | 12V control lead                              | Manual alternator-enable signal only                         | F-15 (3A)                                                         | 16 AWG TXL/GXL                                                               | 6 ft (ASSUMED)                                                    |
| C-42    | 12V panel -> Kicker 46KMC2 media center/source unit                                                          | 12V                                           | Source/head-unit branch (15A max; KMC2 manual shows 15A ATM) | F-10/AUDIO-HU 15A branch plus KMC2 harness fuse                   | 12 AWG duplex if kept near 5 ft; use 10 AWG if longer                        | 5 ft (ASSUMED, driver-side DC shelf)                              |
| C-43    | 12V source/main + stud -> AUDIO-SUB source fuse -> Kicker 49PTRTP10 powered sub +                            | 12V                                           | Powered sub branch; PTRTP10 manual external fuse value       | AUDIO-SUB 40A                                                     | 4 AWG tinned/OFC                                                             | 8 ft (ASSUMED; shorten by placing sub near 12V junction)          |
| C-44    | Kicker 49PTRTP10 powered sub - -> 12V negative bus/main - stud                                               | 12V                                           | Powered sub return                                           | AUDIO-SUB protects paired positive                                | 4 AWG tinned/OFC                                                             | 8 ft (ASSUMED, paired with positive)                              |
| C-45    | KMC2 remote turn-on output -> PTRTP10 remote input                                                           | 12V low-current control                       | Remote amp/sub enable                                        | KMC2/source branch protected                                      | 18 AWG                                                                       | 8 ft (ASSUMED)                                                    |
| C-46    | KMC2 RCA line-out -> PTRTP10 RCA input                                                                       | low-level audio signal                        | Subwoofer signal                                             | N/A                                                               | shielded 2-channel RCA                                                       | measured route (~4m planning cable)                               |
| C-47L/R | KMC2 front speaker outputs -> left/right Kicker CSC67 speakers                                               | speaker-level audio                           | One 4 ohm speaker per KMC2 channel                           | KMC2 internal protection / AUDIO-HU source branch                 | 16 AWG marine speaker wire                                                   | 10-12 ft each (ASSUMED)                                           |
| C-48    | KUS SSS/SSL sender black signal + pink return -> matching signal/negative pair in Cerbo GX MK2 Tank 1 column | Cerbo low-current resistive-sender excitation | Fresh-water level, US 240-30 ohm                             | No external fuse or power feed; do not share chassis return       | 18-22 AWG tinned duplex, ferrules at Cerbo, sealed pigtail splices at sender | Measure after tank/Cerbo endpoints are installed (ASSUMED <10 ft) |
| C-49    | 12V panel -> second Galley USB PD station                                                                    | 12V                                           | Third independent PD branch                                  | F-10 branch fuse (15A); exact panel slot/label to verify in field | 12 AWG duplex                                                                | Owner reports routed 2026-08-05; measure/record final length      |

## Wiring Validation Worksheet (Estimate Pass, 2026-02-18)

Calculation basis for drop screening:

- $V_{\text{drop}} = I * (2 * L_{\text{one\_way}} * R_{\text{per\_ft}})$
- Resistance basis used in this pass ( ohm/ft ): 2/0=0.0000779 , 6 AWG=0.0003951 , 4 AWG=0.0002485 , 12 AWG=0.001588 , 14 AWG=0.002525 , 18/2=0.006385 , 10 AWG=0.000999 .

3. Design targets: <=2% on major 48V trunks, <=3% on planned 12V /AC branches.
4. C-05 is a rollup row that includes both branch and trunk return paths; voltage-drop screen shown is the conservative interim worst-case ( 155A ) within that rollup.

| Circuit ID | From               | To                                    | Fuse             | Current basis                                                                                        | Gauge   | Estimated one-way length                                                                                  | Voltage drop %                      | BC     |
|------------|--------------------|---------------------------------------|------------------|------------------------------------------------------------------------------------------------------|---------|-----------------------------------------------------------------------------------------------------------|-------------------------------------|--------|
| C-01       | Battery A +        | F-01A                                 | F-01A 200A       | 77.5A interim design branch share                                                                    | 2/0 AWG | 2.5 ft planning placeholder; field layout may use unequal positive lengths if total loop path is balanced | 0.059% @ 51.2V planning placeholder | Ro red |
| C-02       | Battery B +        | F-01B                                 | F-01B 200A       | 77.5A interim design branch share                                                                    | 2/0 AWG | 2.5 ft planning placeholder; field layout may use unequal positive lengths if total loop path is balanced | 0.059% @ 51.2V planning placeholder | Ro red |
| C-02C      | Battery C +        | F-01C                                 | F-01C 200A       | 77.5A interim design branch share                                                                    | 2/0 AWG | 2.5 ft planning placeholder; field layout may use unequal positive lengths if total loop path is balanced | 0.059% @ 51.2V planning placeholder | Ro red |
| C-03       | Class T load studs | 48V + bus / disconnect input          | F-01A/B/C        | 77.5A interim per branch                                                                             | 2/0 AWG | 2.5 ft each (x4 conductors)                                                                               | 0.059% @ 51.2V                      | Ro red |
| C-04       | Disconnect output  | Lynx + bus                            | Upstream Class T | 155A interim / 145A final aggregate                                                                  | 2/0 AWG | 2.5 ft                                                                                                    | 0.12% @ 51.2V interim               | Ro red |
| C-05       | Battery - branches | SmartShunt battery side via 48V - bus | N/A              | 77.5A per battery-negative branch; row rollup also includes one 155A interim trunk (NEGBUS_TO_SHUNT) | 2/0 AWG | 2.5 ft each (x4 conductors)                                                                               | 0.12% @ 51.2V (worst-case rollup)   | Ro bla |

|           |                         |                                           |                                        |                                            |                        |              |                            |            |
|-----------|-------------------------|-------------------------------------------|----------------------------------------|--------------------------------------------|------------------------|--------------|----------------------------|------------|
| C-06      | SmartShunt load side    | Lynx - bus                                | N/A                                    | 155A interim / 145A final aggregate return | 2/0 AWG                | 2.5 ft       | 0.12% @ 51.2V interim      | Ro bla     |
| C-06A     | Lynx positive tap       | SmartShunt sense/power lead               | OEM inline fuse                        | OEM harness current                        | OEM harness            | 2.5 ft       | N/A (low-current OEM lead) | Ro ha      |
| C-07      | Lynx Slot 1 DC+         | MultiPlus DC+                             | F-02 125A                              | 125A                                       | 2/0 AWG                | 2.5 ft       | 0.10% @ 51.2V              | Ro red     |
| C-08      | MultiPlus DC-           | Lynx - bus                                | F-02 paired                            | 125A                                       | 2/0 AWG                | 2.5 ft       | 0.10% @ 51.2V              | Ro bla     |
| C-09      | MPPT BAT+               | Lynx Slot 2                               | F-03 60A (80V Victron row 188)         | 45A                                        | 6 AWG                  | 2.5 ft       | 0.17% @ 51.2V              | Ro red     |
| C-10      | MPPT BAT-               | Lynx - bus                                | F-03 paired                            | 45A                                        | 6 AWG                  | 2.5 ft       | 0.17% @ 51.2V              | Ro bla     |
| C-11      | Secondary alternator B+ | Lynx Slot 3 via APM-48                    | F-04 150A                              | 150A design                                | 2/0 AWG                | 20 ft        | 0.80% @ 58.4V              | Ro red     |
| C-12      | Secondary alternator B- | Lynx - bus (dedicated return)             | F-04 paired                            | 150A design                                | 2/0 AWG                | 20 ft        | 0.80% @ 58.4V              | Ro bla     |
| C-13/C-14 | Lynx Slot 4             | Orion 48V +                               | F-05 40A MEGA, >=58VDC                 | 40A fuse basis                             | Existing 6 AWG direct  | 2.5 ft       | 0.16% @ 51.2V              | Ro red     |
| C-15      | Orion 48V -             | Lynx - bus                                | Orion input positive protection paired | 40A fuse basis                             | 6 AWG                  | 2.5 ft       | 0.16% @ 51.2V              | Ro bla     |
| C-18      | Orion 12V +             | Fuse block main + stud                    | F-07 60A/80V Victron row 188           | 30A                                        | 6 AWG                  | 2.5 ft       | 0.49% @ 12V                | Ro red     |
| C-19      | Orion 12V -             | Fuse block integrated - bus / main - stud | F-07 paired                            | 30A                                        | 6 AWG                  | 2.5 ft       | 0.49% @ 12V                | Ro bla     |
| C-19A     | Buffer battery +        | Fuse block main + stud (via F-11/SW)      | F-11 100A                              | 50A design                                 | 4 AWG                  | 2.5 ft       | 0.52% @ 12V                | Ro red     |
| C-19B     | Buffer battery -        | Fuse block integrated - bus / main - stud | N/A                                    | 50A design                                 | 4 AWG                  | 2.5 ft       | 0.52% @ 12V                | Ro bla     |
| C-20      | 12V fuse panel          | Future Starlink direct-DC conversion      | None active                            | N/A                                        | No conductor committed | N/A          | N/A                        | Co BO      |
| C-21      | 12V fuse panel          | Fridge                                    | F-10 15A                               | 7A                                         | 12 AWG duplex          | 12 ft        | 2.23% @ 12V                | Ro fie dup |
| C-22      | 12V fuse panel          | LF Bros diesel-heater harness             | F-10 15A + supplied                    | 10A startup screen                         | 12 AWG duplex          | 8 ft assumed | 2.12% @ 12V                | Fie dup    |

|      |                                |                                                  | harness fuse<br>if present   |                       |                                     |                                        |                                              |                   |
|------|--------------------------------|--------------------------------------------------|------------------------------|-----------------------|-------------------------------------|----------------------------------------|----------------------------------------------|-------------------|
| C-23 | 12V fuse panel                 | Water pump                                       | F-10 10A                     | 7A                    | 14 AWG duplex                       | 8 ft                                   | 2.36% @ 12V                                  | Ro<br>AWG         |
| C-24 | 12V fuse panel                 | CO+propane detector                              | F-10 3A                      | 0.2A                  | 18/2                                | 8 ft                                   | 0.17% @ 12V                                  | Ro                |
| C-25 | 12V fuse panel                 | LED lights +<br>dimmer (Hiatus<br>pre-installed) | F-10 5A                      | 5A                    | 18/2                                | 8 ft                                   | 4.26% @ 12V                                  | Ro                |
| C-26 | 48V system feed                | Cerbo GX                                         | CERB0-PWR 1A-<br>3A          | ~3W                   | 18 AWG duplex<br>acceptable         | 2.5 ft                                 | N/A (low-<br>current<br>electronics<br>feed) | Low<br>ins<br>rov |
| C-27 | PV<br>strings/combiner         | MPPT PV input                                    | F-09A/B/C 15A                | 30A trunk screen      | 10 AWG PV                           | 12 ft trunk<br>+ 3x8 ft<br>string legs | 0.72% @<br>100V trunk<br>screen              | Ro<br>AWG<br>eq   |
| C-28 | Shore inlet path               | AC input<br>breaker/disconnect                   | Source-<br>limited AC<br>OCP | 30A hardware<br>basis | 10/3                                | 8 ft                                   | 0.40% @<br>120VAC                            | Ro<br>sho<br>in/  |
| C-29 | AC input<br>breaker/disconnect | MultiPlus AC-in                                  | Upstream<br>AC OCP           | 30A hardware<br>basis | 10 AWG AC                           | 2.5 ft                                 | 0.12% @<br>120VAC                            | Ro<br>sho<br>in/  |
| C-30 | MultiPlus AC-out-1             | AC-out main<br>breaker                           | 30A AC-out<br>main           | 30A hardware<br>basis | 10/3                                | 2.5 ft                                 | 0.12% @<br>120VAC                            | Ro<br>sho<br>in/  |
| C-31 | Branch A                       | Office/driver<br>receptacle chain                | 20A branch<br>OCP            | 20A                   | 12 AWG AC                           | 15 ft                                  | 0.79% @<br>120VAC                            | Ro<br>AWG         |
| C-32 | Branch B                       | Galley/passenger<br>receptacle chain             | 20A branch<br>OCP            | 20A                   | 12 AWG AC                           | 15 ft                                  | 0.79% @<br>120VAC                            | Ro<br>AWG         |
| C-33 | MultiPlus AC-out-2             | Reserve-only<br>capped route                     | N/A in Phase<br>1            | N/A                   | 12 AWG AC<br>(reserve<br>path only) | 15 ft                                  | 0.60% @<br>120VAC<br>(future-use<br>screen)  | N/A<br>pro<br>Ph  |
| C-34 | 12V fuse panel                 | Office USB PD<br>station                         | F-10 15A                     | 15A design cap        | 12 AWG duplex                       | 5 ft                                   | 1.99% @ 12V                                  | Ro<br>AWG<br>bra  |
| C-35 | 12V fuse panel                 | Galley USB PD<br>station                         | F-10 15A                     | 8A expected           | 12 AWG duplex                       | 8 ft                                   | 1.69% @ 12V                                  | Ro<br>AWG<br>bra  |
| C-49 | 12V fuse panel                 | Second Galley USB<br>PD station                  | F-10 15A; slot<br>to verify  | 15A design cap        | 12 AWG duplex                       | Field<br>routed;<br>measure            | Recalculate<br>from<br>measured<br>length    | Fie               |
| C-36 | 12V fuse panel                 | Maxxair fan<br>(Hiatus pre-<br>installed)        | F-10 10A                     | 4A expected           | 14 AWG duplex                       | 8 ft                                   | 1.35% @ 12V                                  | Ro<br>AWG         |
| C-37 | 12V fuse panel                 | DC<br>ambient/cabinet                            | F-10 5A                      | 5A design cap         | 18/2                                | 8 ft                                   | 4.26% @ 12V                                  | Ro                |

|           |                                        |                                            |                                                 |                                                                   |                                  |                |                       |            |
|-----------|----------------------------------------|--------------------------------------------|-------------------------------------------------|-------------------------------------------------------------------|----------------------------------|----------------|-----------------------|------------|
|           |                                        | LED strips<br>(planned Govee)              |                                                 |                                                                   |                                  |                |                       |            |
| C-38      | House/main positive bus                | WS500 PH-VAN short red lead                | F-12/F-13-PHVAN 15A                             | combined regulator-power / voltage-sense feed at 48v bank voltage | Short harness lead               | Local          | N/A (harness-limited) | Act 171    |
| C-39      | Retired separate positive-sense source | Not installed                              | None                                            | superseded by confirmed PH-VAN combined lead                      | N/A                              | N/A            | N/A                   | Ina 326    |
| C-40      | WS500 current-sense high/low pair      | WS500 current-sense input                  | No fuse per current manual                      | low-current sense; twist pair if extended                         | Harness lead                     | 8 ft           | N/A (harness-limited) | Ha         |
| C-41      | Ford Upfitter #3 output                | WS500 brown ignition/enable input via F-15 | F-15 3A                                         | manual low-current control only                                   | 16 AWG TXL/GXL                   | 6 ft           | N/A (control circuit) | Ro (up cor |
| C-42      | 12V panel                              | Kicker 46KMC2 media center                 | AUDIO-HU 15A branch + KMC2 15A ATM harness fuse | 15A max fuse basis                                                | 12 AWG duplex                    | 5 ft           | 1.99% @ 12V           | Ro au      |
| C-43/C-44 | 12V source/main studs                  | Kicker 49PTRTP10 powered sub               | AUDIO-SUB 40A external source fuse              | 40A manual fuse basis                                             | 4 AWG positive + matching return | 8 ft           | 1.33% @ 12V           | Ro au kit  |
| C-45      | KMC2 remote output                     | PTRTP10 remote input                       | KMC2/source branch protected                    | low-current remote                                                | 18 AWG                           | 8 ft           | N/A                   | Ro au sig  |
| C-46      | KMC2 RCA line-out                      | PTRTP10 RCA input                          | N/A                                             | low-level signal                                                  | shielded RCA                     | measured route | N/A                   | Ro au sig  |
| C-47L/R   | KMC2 front speaker outputs             | Left/right CSC67 speakers                  | KMC2/source branch protected                    | one 4 ohm speaker per channel                                     | 16 AWG marine speaker wire       | 10-12 ft each  | N/A                   | Ro au spe  |

## Wire Rollup (No-Padding Purchase Baseline)

| Gauge / cable family | Estimated total                                            | Source circuits                           | BOM row |
|----------------------|------------------------------------------------------------|-------------------------------------------|---------|
| 2/0 AWG red          | 42.5 ft                                                    | C-01, C-02, C-02C, C-03, C-04, C-07, C-11 | 28      |
| 2/0 AWG black        | 35.0 ft                                                    | C-05, C-06, C-08, C-12                    | 28      |
| 6 AWG red            | 7.5 ft                                                     | C-09, C-13/C-14, C-18                     | 29      |
| 6 AWG black          | 7.5 ft                                                     | C-10, C-15, C-19                          | 29      |
| 4 AWG red            | 10.5 ft (2.5 ft existing + 8 ft audio sub planning branch) | C-19A, C-43                               | 30, 192 |
| 4 AWG black          | 10.5 ft (2.5 ft existing + 8 ft audio sub return)          | C-19B, C-44                               | 30, 192 |

|                                            |                                                                                                                                      |                                                                                                                                               |                        |
|--------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------|------------------------|
| 10 AWG pair-equivalent (PV)                | placeholder only                                                                                                                     | C-27 final module/string topology deferred until solar workstream; do not buy combiner/fuse count or cut roof entries from the old 3S3P model | 31                     |
| 14 AWG duplex                              | 16 ft                                                                                                                                | C-23, C-36                                                                                                                                    | 32                     |
| 18/2                                       | 24.0 ft plus separate Cerbo low-current duplex                                                                                       | C-24, C-25, C-37; Cerbo C-26 uses 48V fused low-current duplex                                                                                | 33 / low-current stock |
| 12 AWG AC branch cable                     | 30 ft (C-33 excluded in Phase 1)                                                                                                     | C-31, C-32                                                                                                                                    | 113                    |
| 10/3 shore + AC-in/out feed                | 13 ft                                                                                                                                | C-28, C-29, C-30                                                                                                                              | 114                    |
| 12 AWG fridge/heater/USB/audio branch mix  | Fridge: 12 ft; heater: 8 ft; USB: 5 ft office + 8 ft Galley A + measured Galley B field run; audio source: 5 ft short-run assumption | C-21, C-22, C-34, C-35, C-49, C-42                                                                                                            | 116, 193 / field stock |
| Camper audio signal/speaker/control wiring | RCA measured route, 18 AWG remote, and 16 AWG marine speaker wire runs                                                               | C-45, C-46, C-47L/R                                                                                                                           | 193                    |
| WS500 harness/sense leads                  | Included in selected kit/harness set; C-39 is retired under PH-VAN                                                                   | C-38, C-40                                                                                                                                    | 168, 171               |
| 16 AWG TXL/GXL control wire                | 6 ft                                                                                                                                 | C-41                                                                                                                                          | 176                    |

Notes:

1. This table is a base estimate only; it intentionally excludes order padding and termination waste.
2. Apply personal order overage at checkout based on actual spool cut increments and routing confidence.
3. Parallel bank balancing is locked by similar total loop resistance per battery. Positive-only leads may differ if the paired negative path offsets the difference; verify with final measured lengths and, if needed, clamp-current checks.

### 3x Battery Bank Bench-Build Cut List (2/0 AWG)

Purpose: make the bench build orderable without needing final camper run lengths. Treat lengths below as *bench module* lengths only; final install harnesses should be re-cut after layout freeze.

Assumptions:

1. Battery terminals are M8 (verify your battery stud size before ordering lugs).
2. Class T fuse blocks and battery-side busbars use 3/8" studs (treat as M10 lugs unless your specific hardware differs).
3. Lynx Distributor is the M10 model (main connections M10 ; internal/fuse studs may still be M8 depending on the position).

| Cable ID             | Qty | From -> To                              | Color | Gauge | Est. one-way length   | Lug A | Lug B |
|----------------------|-----|-----------------------------------------|-------|-------|-----------------------|-------|-------|
| BATT+_A/B/C          | 3   | Battery + -> Class T block line side    | red   | 2/0   | 2.5 ft each (ASSUMED) | M8    | M10   |
| FUSE_TO_POSBUS_A/B/C | 3   | Class T block load side -> 48V + busbar | red   | 2/0   | 2.5 ft each (ASSUMED) | M10   | M10   |
| POSBUS_TO_DISC       | 1   | 48V + busbar -> disconnect input        | red   | 2/0   | 2.5 ft (ASSUMED)      | M10   | M10   |
| DISC_TO_LYNX+        | 1   | disconnect output -> Lynx + input       | red   | 2/0   | 2.5 ft (ASSUMED)      | M10   | M10   |
| BATT-_A/B/C          | 3   | Battery - -> 48V - busbar               | black | 2/0   | 2.5 ft each (ASSUMED) | M8    | M10   |
| NEGBUS_TO_SHUNT      | 1   | 48V - busbar -> SmartShunt battery side | black | 2/0   | 2.5 ft (ASSUMED)      | M10   | M10   |
| SHUNT_TO_LYNX-       | 1   | SmartShunt load side -> Lynx - input    | black | 2/0   | 2.5 ft (ASSUMED)      | M10   | M10   |
| LYNX_SLOT1_TO_MULTI+ | 1   | Lynx Slot 1 DC+ -> MultiPlus DC+        | red   | 2/0   | 2.5 ft (ASSUMED)      | M8    | M8    |
| LYNX_TO_MULTI-       | 1   | Lynx - -> MultiPlus DC-                 | black | 2/0   | 2.5 ft (ASSUMED)      | M8    | M8    |



Locked balancing rule for the 3x parallel bank:

1. Keep each battery path's total positive + negative loop resistance similar.
2. Equal positive-only leads are not required when the negative lead lengths intentionally offset the positive lead differences.
3. Keep the same cable family, lug geometry, fuse/holder family, and termination quality across all three battery paths.
4. Verify sharing with clamp-current checks under charge/load if final measured path lengths differ materially.

Torque reference (verify against your exact manuals/hardware):

- MultiPlus-II DC terminals: 12 Nm ( M8 nut) per Victron installation guidance.
- SmartShunt shunt bolts: verify torque for the installed 300A model per Victron installation guidance.
- Lynx Distributor M10 model: M10 nuts 33 Nm (older serials may be lower), and M8 nuts 14 Nm per Victron Lynx installation guidance.

## Additional Components Included In Topology Scope

- 48V disconnect ( 275A )
- Pre-charge resistor (commissioning/soft-charge aid before connecting large DC loads)
- Battery-side 48V + combine busbar (after Class T fuses)
- Battery-side 48V - combine busbar (battery-only, before SmartShunt)
- 12V fuse block main + stud used as source-combine point (Orion + buffer battery feed)
- 12V fuse block integrated negative bus/main - used as shared return point
- 12V buffer battery main fuse ( F-11 ) and manual disconnect switch ( SW-12V-BATT )
- Shore AC inlet + cord/adaptor interface hardware
- Portable EMS in source-side shore path before camper inlet
- Single combined 6-way AC DIN enclosure with isolated AC-in and AC-out neutral paths plus common equipment grounding/PE handling
- AC-out 30A main breaker plus 20A / 20A branch breaker and GFCI receptacle hardware
- Receptacle boxes + 120V outlets (current purchased baseline 2 GFCI receptacles/covers in row 15 ; inactive row 112 records the closed separate box/faceplate allowance)
- AC-out-2 reserve-only capped route (no energized Phase 1 branch hardware)
- USB PD station branch hardware ( 3 stations / independent routes owner-reported 2026-08-05 : 1x office and 2x Galley; row 115 owns the exact purchased Acegoo evidence while the additional station purchase provenance remains untracked)
- Camper audio hardware: Kicker 46KMC2 source branch, Kicker 49PTRTP10 powered sub branch, AUDIO-HU 15A source protection, AUDIO-SUB 40A source fuse, RCA/speaker/remote wiring
- Battery temperature sensor wiring to inverter/monitoring path
- SmartShunt fused positive sense/power lead (factory harness)
- Cerbo GX CERB0-PWR small inline fused 48V power feed
- Ford Upfitter #3 control lead and local F-15 inline fuse for the WS500 brown ignition/enable wire

## Assumptions (Explicit)

1. Cable sizing assumes fine-strand copper conductors (OFC welding-cable baseline for high-current DC paths), enclosed vehicle routing, and the estimated one-way lengths listed in this document.
2. Voltage-drop design intent used here:  $\leq 2\%$  on major 48V power runs and  $\leq 3\%$  on 12V branch circuits.
3. F-09 PV string fuse value ( 15A ) remains provisional until final module datasheet max-series-fuse rating is confirmed.
4. Cerbo GX feed is now a small inline fused 48V feed ( CERB0-PWR ) from the system/load side of the main disconnect during bench commissioning, so the Cerbo powers down with the house system.
5. Orion branch uses one source-side fuse only: F-05 40A MEGA, body-marked at least 58VDC , in Lynx Slot 4 feeding the existing 6 AWG input pair directly. The 58VDC minimum is tied to the locked 56.8V charge ceiling; use Victron CIP138040020 40A/80V if replacement is needed. Standalone F-06 is retired; do not add an inline fuse after the Lynx slot.
6. No low-voltage-disconnect (LVD) automation is included in Phase 1; protection is source fusing plus manual SW-12V-BATT isolation.
7. Alternator architecture lock is dedicated 48V secondary alternator path ( Mechman + WS500 + APM-48 ) with F-04 150A ; obsolete pre-Mechman engine-bay fuse paths are removed from active layout.
8. F-01A/B/C are provisionally set to 200A pending final 51.2V battery datasheet/manual confirmation; if validated limits are lower, shift to 175A .
9. 2/0 cable quantity planning baseline in this pass is 77.5 ft total no-padding ( 42.5 ft red + 35.0 ft black); user-applied order padding is intentionally deferred to checkout.
10. Manual alternator shutdown baseline is Ford Upfitter #3 feeding the WS500 brown ignition/enable wire through F-15 ; WS500 white Feature-In remains a future-only reserve for automatic interlock work.
11. Camper audio is a 12V accessory load. KMC2 source branch is 15A ; PTRTP10 sub branch is 40A with 4 AWG power and return. Audio peaks can exceed Orion continuous output and rely on the 12V buffer battery, so first loud testing should watch 12V voltage/SOC.

## Completion Status

- DC/PV topology is complete for current BOM scope and load model scope.
- AC hierarchy is complete at architecture level, including transfer behavior, branch strategy, and protection chain.
- Full-circuit estimate pass is now documented with run lengths, voltage-drop screening, and purchase rollups.
- Camper audio branch added as a DC-first 12V accessory package: Kicker 46KMC2 source branch plus Kicker 49PTRTP10 powered sub branch; detailed product/routing notes live in docs/implementation/CAMPER\_audio\_system.md .
- Remaining work is final measured-length replacement and SKU-level closeout before final cut-to-length harness production.